AN INTEGRAL MORPHISM HAS EXACT PUSHFORWARD ON ETALE SHEAVES AND VANISHING HIGHER DIRECT IMAGES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-spaces-cohomology/preamble.tex is excerpted verbatim as three spans, lines 8-12 (verbatim package and the project's own reference/slogan/history display environments), lines 14-18 (Xy-pic setup) and lines 154-217 (theorem declarations and mathematical macros). The wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler. No macro was invented or redefined.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Higher direct images can be computed after an etale cover of the target. Properties of Spaces, label spaces-properties-lemma-pushforward-etale-base-change (invoked at proof line 505). It reduces the statement to the case where Y is a scheme.
- An integral morphism to a scheme has a scheme as source. Morphisms of Spaces, label spaces-morphisms-lemma-integral-local (invoked at proof line 508). Together with the reduction it puts both spaces in the realm of schemes, where the etale cohomology statement is available.
- Exactness of the pushforward and vanishing of higher direct images for integral morphisms in etale cohomology. Etale Cohomology, labels etale-cohomology-lemma-what-integral (invoked at proof line 511) and etale-cohomology-proposition-finite-higher-direct-image-zero (invoked at proof line 513). The first settles the integral case; the second is the less technical reference for the finite case that the author points the reader to.

Source reference. Original lemma 04C2: sources/stacks-chow/tags.txt, source lines 5676–5676. Navigation pointer for \ref{etale-cohomology-lemma-what-integral} used inside the counted span (line 511); the argument that consumes it is retained in the span.

Source reference. Original proposition 03QP: sources/stacks-chow/tags.txt, source lines 4902–4902. Navigation pointer for \ref{etale-cohomology-proposition-finite-higher-direct-image-zero} used inside the counted span (line 514); the argument that consumes it is retained in the span.

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Source reference. Original lemma 03ZQ: sources/stacks-chow/tags.txt, source lines 5229–5229. Navigation pointer for \ref{spaces-morphisms-lemma-integral-local} used inside the counted span (line 509); the argument that consumes it is retained in the span.

Source reference. Original lemma 03LR: sources/stacks-chow/tags.txt, source lines 4723–4723. Navigation pointer for \ref{spaces-properties-lemma-pushforward-etale-base-change} used inside the counted span (line 505); the argument that consumes it is retained in the span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be an integral (for example finite) morphism of algebraic spaces. Then $f_* : Ab(X_{étale}) \to Ab(Y_{étale})$ is an exact functor and $R^p f_* = 0$ for $p > 0$.*

Proof. By Properties of Spaces, Lemma 03LR we may compute the higher direct images on an étale cover of Y. Hence we may assume Y is a scheme. This implies that X is a scheme (Morphisms of Spaces, Lemma 03ZQ). In this case we may apply Étale Cohomology, Lemma 04C2. For the finite case the reader may wish to consult the less technical Étale Cohomology, Proposition 03QP. □